

MkrPOS++

Next-Generation Multi-Tenant POS & Unified Retail Management Suite

A premium, high-performance, and secure full-stack software solution designed for modern retail outlets, multi-store brands, and SaaS hosting providers.

Decoupled Performance

React 18 frontend paired with Native PHP backend for sub-500ms checkout times.

Multi-Tenant Architecture

Strict database-level data isolation for secure multi-outlet or SaaS deployment.

Advanced Cashier Controls

Multi-tab sales, barcode scanning, draft billing, and thermal printer integration.

Unified Operations

Built-in stock adjustments, wastage write-offs, costs audit, and customer loyalty.

PREPARED FOR

Business Stakeholders & Clients

PREPARED BY

Engineering & Product Team

DATE

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1. Introduction & Project Scope

In today's fast-paced retail ecosystem, merchants struggle with disconnected software platforms. Inventory is managed in one system, billing is done on a localized machine, and customer loyalty or expense records are written down or tracked on spreadsheets. This disconnect leads to stockouts, manual accounting errors, and lost revenue opportunities.

MkrPOS++ is an all-in-one software package built to solve these exact challenges. By merging an ultra-fast billing interface with a centralized back-office administration suite, MkrPOS++ allows businesses to run smoothly. The software has been designed from the ground up for **Multi-Tenant Operations**, making it capable of running multiple separate stores in isolation on a single database.

KEY VALUE METRIC

Unlike heavy, resource-intensive software suites, MkrPOS++ operates on an extremely optimized stack (React + Lightweight Native PHP). This reduces server maintenance costs by up to **80%** and allows local stores to load pages and process sales in under **500ms**.

Core Operating Modes

- **Multi-Branch Corporate POS:** A single company operates several physical branches. The management has global insights into sales, inventory, and staff across all branches, while staff members are isolated to their specific assigned locations.
- **Commercial SaaS Platform:** The operator licenses the software to independent retail store owners. Each shop acts as a "tenant" with its own products, customers, settings, and staff. The operator collects a monthly or annual subscription fee.

2. Product Features & Modules

MkrPOS++ is divided into user-friendly modules that cover all key business tasks:

High-Speed Checkout Terminal

Supports multi-tab active billing, real-time barcode scanner focus, flat/percentage discounts, dynamic tax calculations, and thermal print receipt outputs.

Smart Inventory & Catalog

Tracks items in custom units (kg, liters, pieces), issues automated low-stock warnings, and supports bulk product uploads via secure CSV files.

Loyalty & Customer Profiles

Log customer contact details, track credits/dues, and reward loyalty points based on spend thresholds that customers can redeem on checkouts.

Wastage & Returns Auditing

Register returned stock with automated refund logging and record spoiled, expired, or stolen items (wastage) to maintain accurate stock sheets.

Expense & Operations Control

Log store operating costs like rent, utilities, salaries, and marketing to keep track of store expenditures.

Integrated Profit/Loss Charts

A back-office reporting center that calculates net profit by automatically matching sales revenue against operational costs and COGS.

Staff Permission Matrix

Access controls ensure staff members can only perform checkout and stock checks, while administrators have full access to settings and finances.

Capability / Module	Super Admin	Shop Admin	Shop Staff
Manage Tenants & Shop Licenses	Yes	No	No
Change Global Password Reset Settings	Yes	No	No
Access Checkout Terminal	No	Yes	Yes
Configure Loyalty Rules & Tax Rates	No	Yes	No
Add / Edit Store Staff & Salaries	No	Yes	No
View Financial Profitability & Charts	No	Yes	No
Perform Inventory Returns & Wastage Logs	No	Yes	No

3. Technical Architecture & Security

The architecture of MkrPOS++ is optimized for scalability and data security, keeping maintenance simple for the engineering team.

Stateless Authentication & Session Isolation

Security is managed using JSON Web Tokens (JWT) and dynamic tenant checks:

- **Secure JWTs:** On login, users receive a stateless cryptographic JWT. The client stores this token securely and attaches it to the `Authorization: Bearer` header for all subsequent API requests.
- **Database Tenant Separation:** The database enforces tenant isolation. Every request is filtered by the user's shop ID (`tenant_id`). It is impossible for data from one shop to be visible to another, ensuring total data security.
- **Password Encryption:** Passwords are encrypted using native PHP `bcrypt` algorithms, protecting credential storage from breach attempts.

Data Schema and Architecture Summary

Stack Layer	Technology	Role in System
Client Interface	React 18, Vite, CSS	Fast single-page application, client-side cart operations, and instant layouts.
API Backend	Native PHP 8.2 (PDO)	Handles database queries, security validations, and router functions with zero dependencies.
Database	MySQL 8	Maintains store data, inventories, customer records, and transaction logs.
Integrations	Standard ESC/POS Printing	Provides printable checkout templates for 80mm thermal printers.

ZERO-FRICTION DEPLOYMENT ENGINE

Unlike systems that require complex server setups (like Laravel or Node backend servers), the PHP backend in MkrPOS++ runs on standard cPanel environments, low-cost VPS instances, or web hosting platforms. The system runs automatic database setup scripts on startup.

4. Business Roadmap & Integration

MkrPOS++ is ready to scale. Future updates can extend its capabilities to meet growing business needs:

Integration & Extension Options

- **E-Commerce Synchronization:** Connect your retail store directly to online storefronts (e.g., WooCommerce, Shopify, or a custom React web application). Keep stock counts synchronized across web sales and physical checkout lines.
- **Wholesale Customer Portal:** Enable wholesale accounts to view catalogs, request price quotes, and manage outstanding balances through their own logins.
- **Mobile POS Companion App:** Build on the existing API to launch lightweight Android/iOS apps, allowing staff to manage inventory and scan barcodes directly from mobile devices.
- **Multi-Location Data Hub:** Centralize reporting dashboards for corporate administrators to manage multiple brands from a single login.

Project Handover & SLA Commitment

We deliver the code with clean documentation, clear installation scripts, and testing setups.

Deliverable Phase	Scope of Delivery	Support Timeline
Core Installation	Full backend, frontend single-page application, and database schema scripts setup on client host server.	Weeks 1 - 2
Custom Branding	Integration of client corporate logos, default currencies, localized tax levels, and print receipt footers.	Week 3
Staff Onboarding	Online walkthrough sessions for admin managers, cashiers, and stock controllers.	Week 4
SLA Support	12 months of security patches, software updates, and server monitoring support.	12 Months Support

Seal the Project with MkrPOS++

Deliver an efficient, responsive, and secure experience to cashiers and store admins while keeping server operating costs low.

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